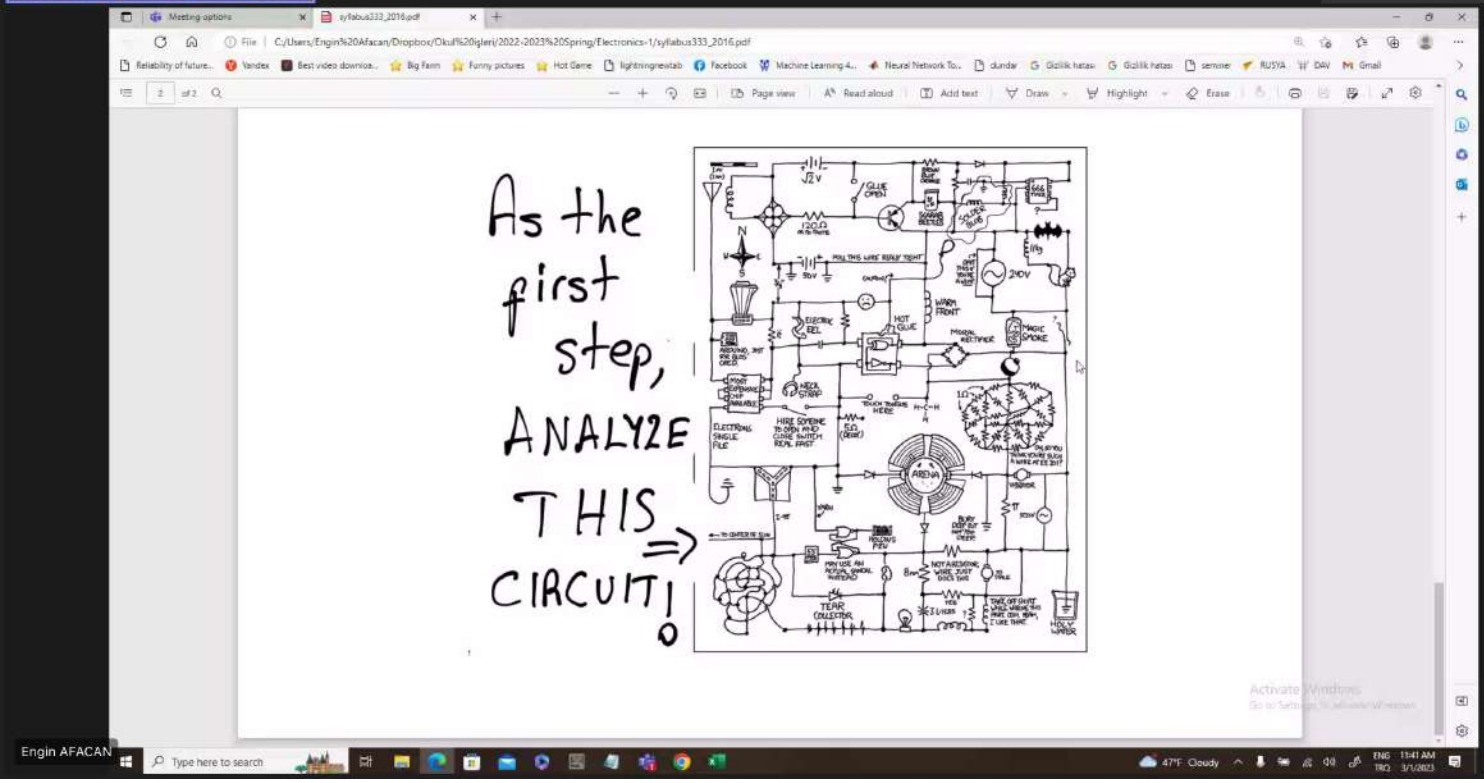
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1.4.7. Amplifier Saturation

- **limited linear range** – practically, amplifier operation is linear over a limited input range.
- **saturation** – beyond this range, saturation occurs.
  - output remains constant as input varies

$$\underbrace{\frac{L_{minus}}{A_v} \le v_i \le \frac{L_{plus}}{A_v}}_{\text{or...}\quad L_{minus} \le v_o \le L_{plus}}$$

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1.4.6. Amplifier Power Supply

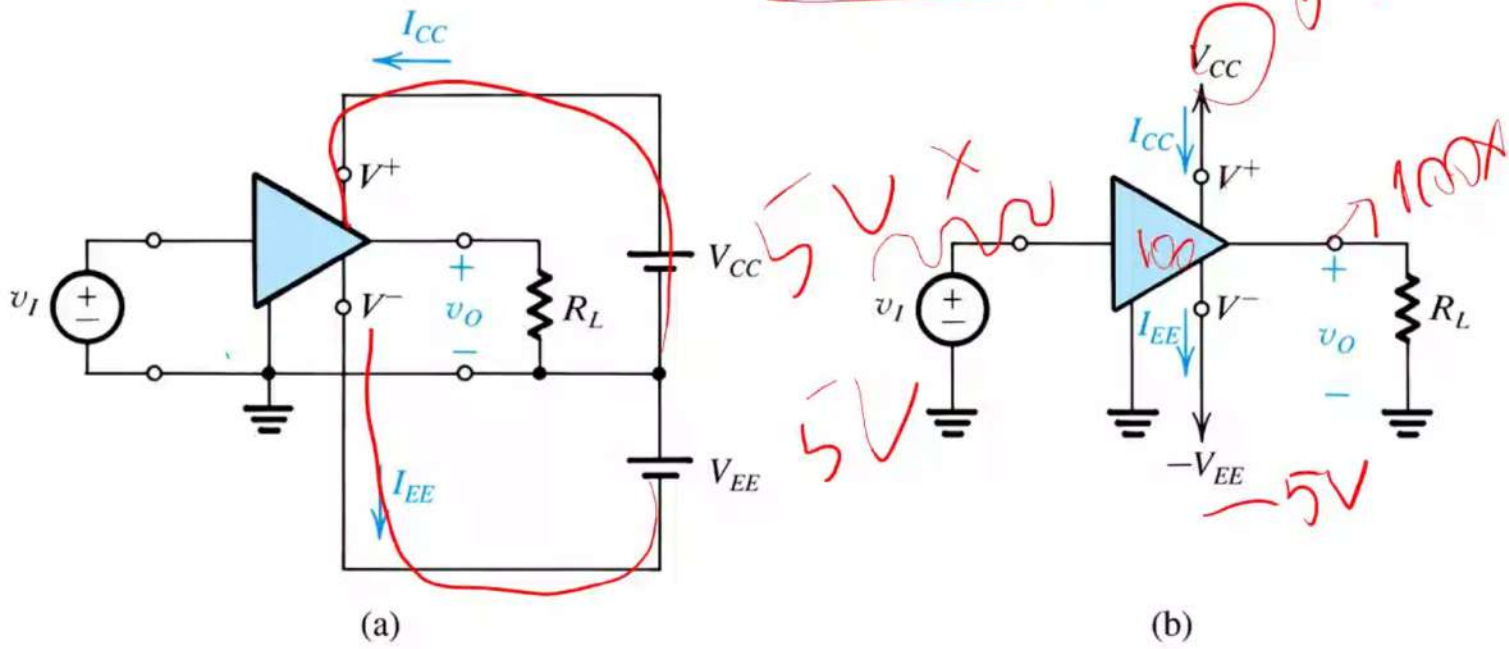


Figure 1.13: An amplifier that requires two dc supplies (shown as batteries) for operation.

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1.4.6. Amplifier Power Supply

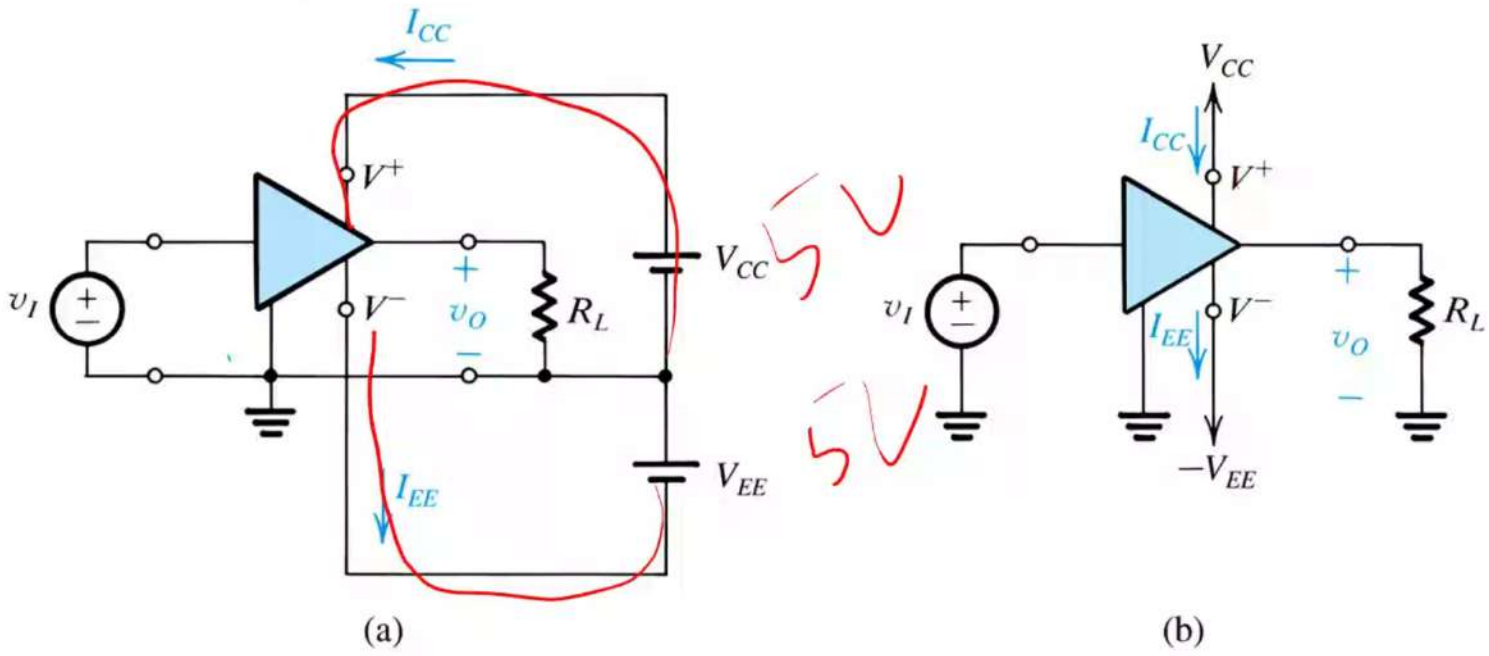


Figure 1.13: An amplifier that requires two dc supplies (shown as batteries) for operation.

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1.4.6. Amplifier Power Supply

- **conservation of power** – dictates that power input (P_i) plus that drawn from supply (P_{dc}) is equal to output (P_L) plus that which is dissipated (P_{dis}).
  - $P_i + P_{dc} = P_L + P_{dissapated}$
- **efficiency** – is the ratio of power output to input.
  - efficiency = $P_L / (P_i + P_{dc})$

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1.4.6. **Amplifier Power Supply**

- **supplies** – an amplifier has two power supplies
  - V_{CC} is positive, current I_{CC} is drawn
  - V_{EE} is negative, current I_{EE} is drawn
- **power draw** – from these supplies is defined below
  - $P_{dc} = V_{CC} I_{CC} + V_{EE} I_{EE}$

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1.4.5. **Expressing Gain in Decibels**

- **Q:** How may gain be expressed in **decibels**?

$$\text{voltage gain in decibels} = 20\ \log|A_v|\ dB$$

$$\text{current gain in decibels} = 20\ \log|A_i|\ dB$$

$$\text{power gain in decibels} = 10\ \log(A_p)\ dB$$

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1.4.4. Power and Current Gain

- **Q:** What is one main difference between an amplifier and transformer? …Because both alter voltage levels.
  - **A:** Amplifier may be used to boost power delivery.

$$power\ gain\ (A_p) = \frac{load\ power\ (P_L)}{input\ power\ (P_I)} = \frac{v_o i_o}{v_i i_i}$$

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1.4.2. Amplifier Circuit Symbol

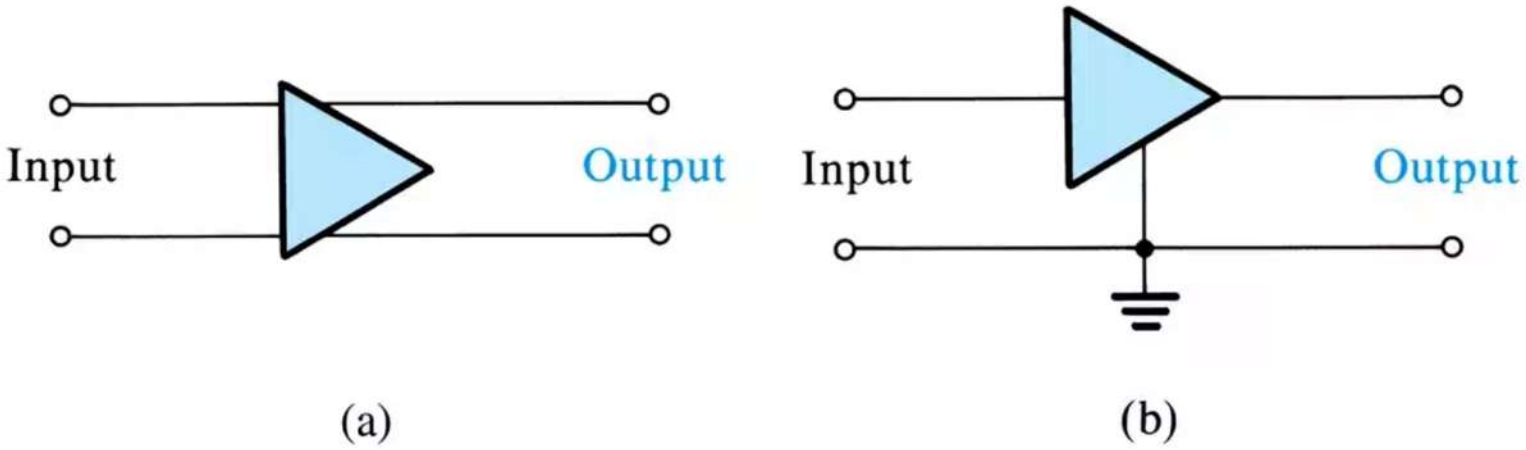


Figure 1.11: (a) Circuit symbol for amplifier. **(b)** An amplifier with a common terminal (ground) between the input and output ports.

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1.4.1. Signal Amplification

- **voltage amplifier** – is used to boost voltage levels for increased resolution.
- **power amplifier** – is used to boost current levels for increased "intensity".

output / input relationship for amplifier

$$\mathbf{v}_o(t) = A_v \mathbf{v}_i(t)$$

voltage gain

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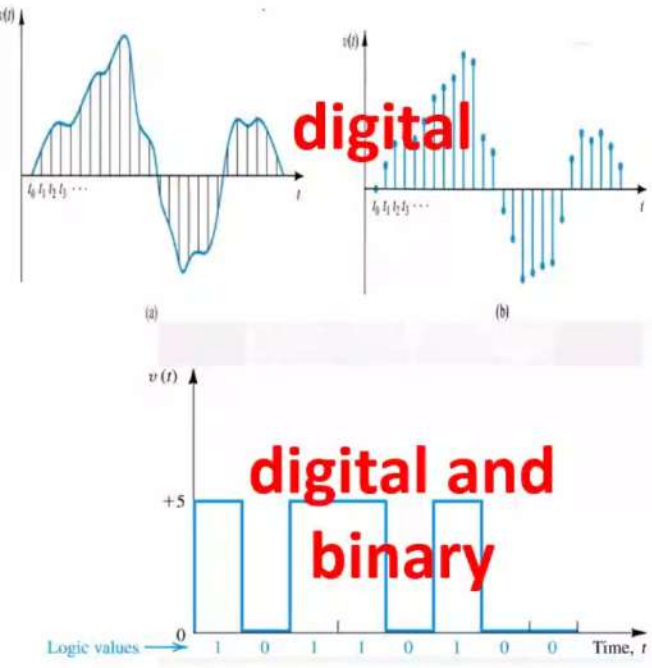
1.4. **Amplifiers**

- **Q:** Why is signal amplification needed?
  - **A:** Because many transducers yield output at low power levels (*mW*)
- **linearity** – is property of an amplifier which ensures a signal is not "altered" from amplification
- **distortion** – is any unintended change in output

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1.3. Analog and Digital Signals



- **Q:** Are digital and binary synonymous?
  - **A:** No. The binary number system (base$_2$) is one way to represent digital signals.

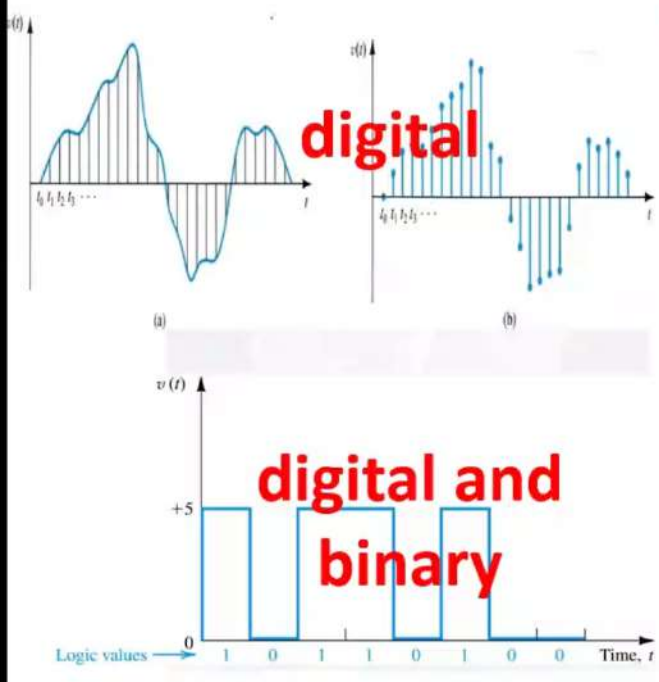
$$\overbrace{y = \underbrace{b_0}_{LSB} 2^0 + b_1 2^1 + b_2 2^2 + \ldots}^{\text{base 10} \leftarrow \text{base 2}}$$

$$\ldots + b_3 2^3 + \ldots \underbrace{b_{n-1}}_{MSB} 2^{n-1}$$

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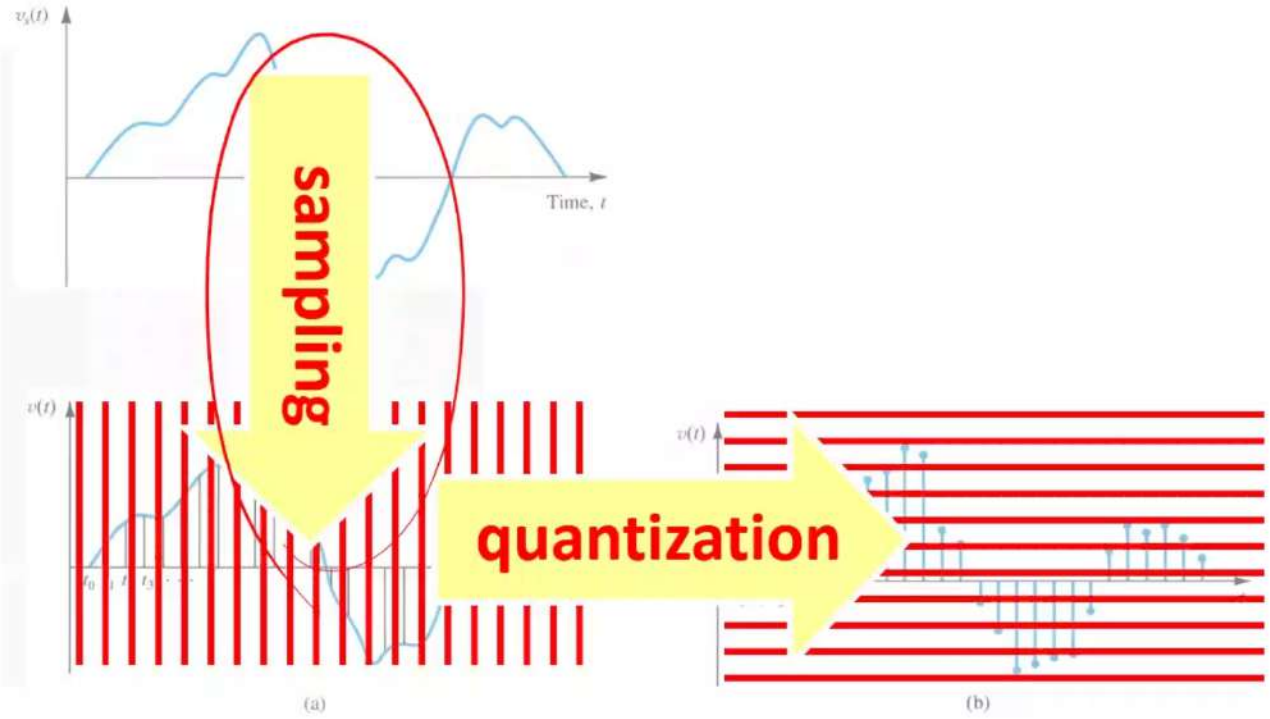
$$y = \underbrace{b_0}_{LSB} 2^0 + b_1 2^1 + b_2 2^2 + \ldots$$

$$\ldots + b_3 2^3 + \ldots \underbrace{b_{n-1}}_{MSB} 2^{n-1}$$

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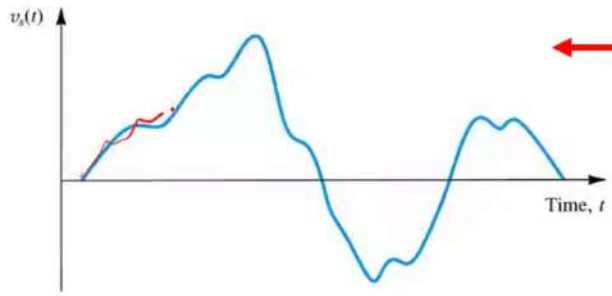
1.3. Analog and Digital Signals



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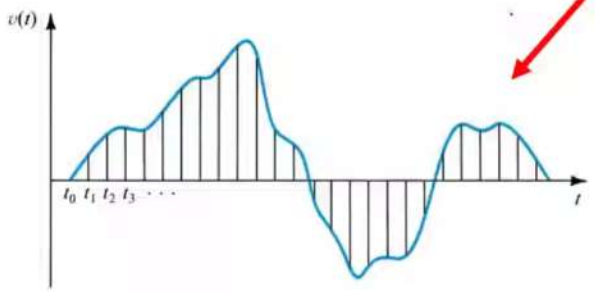
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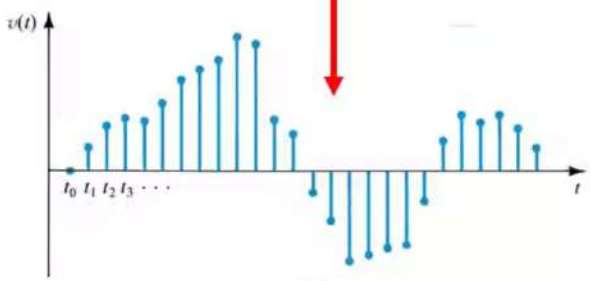
analog signal

discrete-time signal

digital signal



(a)



(b)

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1.3. Analog and Digital Signals

- **analog signal** – is continuous with respect to both value and time
- **discrete-time signal** – is continuous with respect to value but sampled at discrete points in time
- **digital signal** – is quantized (applied to values) as well as sampled at discrete points in time

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1.2. **Frequency Spectrum of Signals**

- **Q:** Can the Fourier Transform be applied to a non-periodic function of time?
  - **A:** Yes, however (as opposed to a discrete frequency spectrum) it will yield a continuous…

